

SKILL WEEK 6

Task 1:

Code:

```
#include <stdio.h>

int main() {
    char input[200];
    int i;

    printf("Enter a string with escape characters: ");
    fgets(input, sizeof(input), stdin);

    printf("\nProcessed string:\n");
    for (i = 0; input[i] != '\0'; i++)
    {
        if (input[i] == '\\' && input[i + 1] != '\0')
        {
            i++;

            switch (input[i])
            {
                case 'n':
                    printf("\n");
                    break;
                case 't':
                    printf("\t");
                    break;
                case ' ':
                    printf(" ");
                    break;
                case '\\':
                    printf("\\");
                    break;
                case '$':
                    printf("$");
                    break;
                case '!':
                    printf("!");
                    break;
```

```

default:
    printf("%c", input[i]);
    }

    } else {
printf("%c", input[i]);
    }
    }

    return 0;
}

```

```

venkat038@Ayyappa:~$ nano 6.1.c
venkat038@Ayyappa:~$ gcc 6.1.c -o 6.1
venkat038@Ayyappa:~$ ./6.1
Enter a string with escape characters: Hello\nWorld

Processed string:
Hello
World
venkat038@Ayyappa:~$ |

```

Task 2:

Code:

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {

    pid_t pid;

    pid = fork();
    if (pid < 0) {
perror("fork failed");
return 1;
    }

    if (pid == 0) {

        // Child process
        printf("Child
process created.\n");
        PID: %d\n", getpid());
        printf("Child

```

```
    char *args[] = {"ls", "-l", NULL};    execvp(args[0],
args);

    // This runs only if execvp fails
    perror("execvp failed");    exit(1);

} else {

    // Parent process
    printf("Parent process.\n");
    printf("Parent PID: %d\n", getpid());

    wait(NULL);

    printf("Child process finished.\n");
}

return 0;
}
```

Output:

```
venkat038@AYYAPPA:~$ nano 6.2.c
venkat038@AYYAPPA:~$ gcc 6.2.c -o 6.2
venkat038@AYYAPPA:~$ ./6.2
Parent process.
Parent PID: 1124
Child process created.
Child PID: 1125
total 992
-rw-r--r-- 1 venkat038 venkat038 98 Jul 31 09:35 1.c
-rw-r--r-- 1 venkat038 venkat038 94 Jul 31 09:43 2.c
-rw-r--r-- 1 venkat038 venkat038 163 Jul 31 09:52 3.c
-rw-r--r-- 1 venkat038 venkat038 345 Jul 31 09:58 4.c
-rwxr-xr-x 1 venkat038 venkat038 16176 Sep 21 14:31 5.1
-rw-r--r-- 1 venkat038 venkat038 385 Sep 21 14:31 5.1.c
-rwxr-xr-x 1 venkat038 venkat038 16216 Sep 21 14:34 5.2
-rw-r--r-- 1 venkat038 venkat038 476 Sep 21 14:34 5.2.c
-rw-r--r-- 1 venkat038 venkat038 262 Aug 11 03:17 5.c
-rwxr-xr-x 1 venkat038 venkat038 16032 Aug 11 03:21 6
-rwxr-xr-x 1 venkat038 venkat038 16176 Sep 21 14:39 6.1
-rw-r--r-- 1 venkat038 venkat038 1042 Sep 21 14:39 6.1.c
-rwxr-xr-x 1 venkat038 venkat038 16296 Sep 21 14:40 6.2
-rw-r--r-- 1 venkat038 venkat038 720 Sep 21 14:40 6.2.c
-rw-r--r-- 1 venkat038 venkat038 131 Aug 11 03:20 6.c
-rwxr-xr-x 1 venkat038 venkat038 16072 Aug 11 03:22 7
-rwxr-xr-x 1 venkat038 venkat038 16248 Sep 21 04:39 7.1
-rw-r--r-- 1 venkat038 venkat038 838 Sep 21 04:39 7.1.c
-rwxr-xr-x 1 venkat038 venkat038 16296 Sep 21 04:40 7.2
```